Module 1:
What are rays?

Course 1 of *Optical Engineering*: First Order Optical System Design

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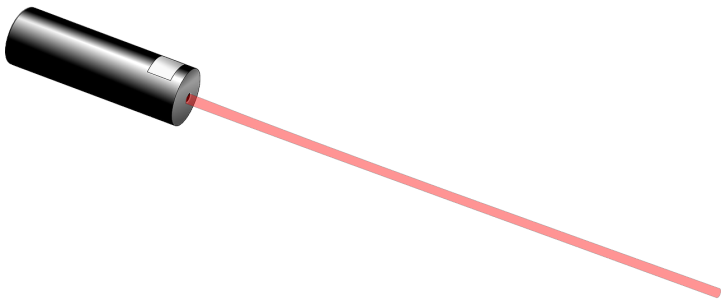
What are rays?



A “pencil beam” of light:
Not wrong, but imprecise

Rays actually describe solutions to the electromagnetic wave equation

$$\nabla^2 E + k^2 E = 0$$

Let the solution be described by an amplitude E(r) and a phase k_0 S(r) where S is the “optical path length” in meters:

$$\vec{E}(\vec{r}) = E(\vec{r}) e^{-jk_0 S(\vec{r})}$$

$$k_0 S(\vec{r}) = k_x x + k_y y + k_z z$$ Plane wave

$$k_0 S(\vec{r}) = k_0 \sqrt{x^2 + y^2 + z^2}$$ Spherical wave

k_0	Wave number of free space	$\omega/c = 2\pi/\lambda_0$	[1/m]
k	Wave number in a material	$n\,k_0 = 2\pi n/\lambda_0$	[1/m]
c	Speed of light in vacuum		[m/s]

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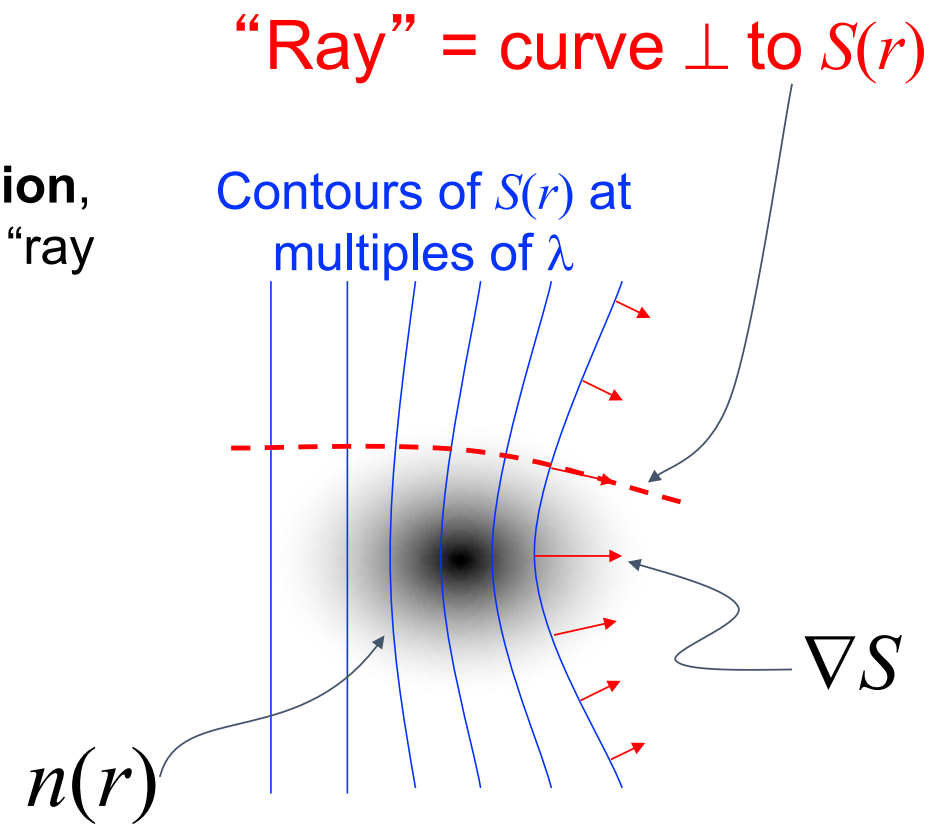
What are rays?

Under the approximation that the **amplitude E changes slowly with position**, the wave equation can be reduced to the "ray equation"

$$\left|\nabla S(\vec{r})\right|^2 = n^2(\vec{r})$$

The optical path length is just the integral of the index along the ray path

$$S(r) = \int_A^B n(\vec{r})ds \qquad \text{[m]}$$



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What are rays?

The "ray equation" can be transformed into the eikonal ("image") equation with no further approximations

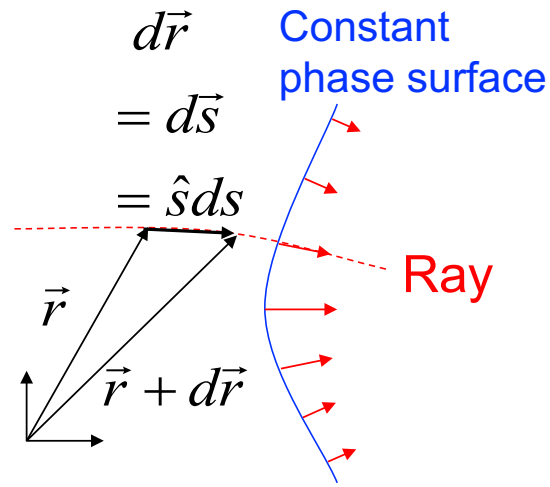
$$\frac{d}{ds}\left[n(\vec{r})\frac{d\vec{r}}{ds}\right]=\nabla n(\vec{r})$$

If the refractive index is a constant

$$\frac{d}{ds}\left[n_0\frac{d\vec{r}}{ds}\right]=\nabla n_0=\vec{0}$$

which has solution

$\vec{r}(s)=\vec{c}\,s$ where $\vec{c}$ is a constant.



Summary

1. Rays are solutions to Maxwell's equations
2. …IF the amplitude of the field varies slowly
3. Rays are normal to the surfaces of constant phase.
4. The integral of n along the ray is the OPL.
5. Rays are straight lines in homogeneous space.

Saleh & Teich 1.3, Born & Wolf 7th edition, pg. 129

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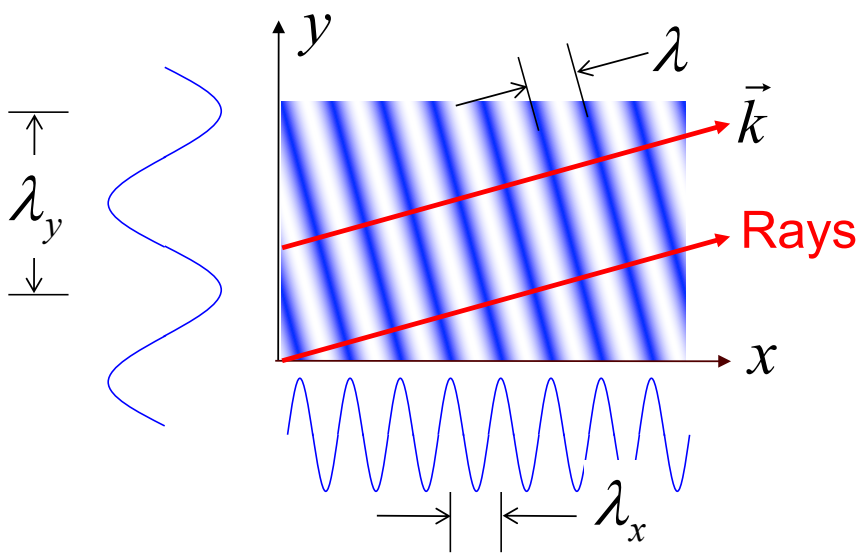
What are rays: Examples

Plane wave

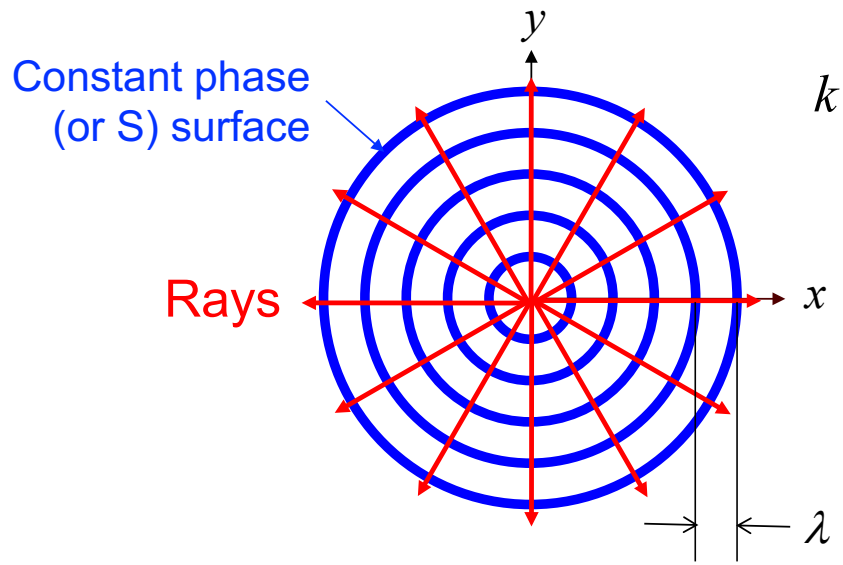
$$\vec{E}(x,y,z,t) = \vec{E}_0 e^{j\left[\omega t - \left(k_x x + k_y y + k_z z\right)\right]}$$



Spherical wave

$$E(x,y,z,t) = E_0 \frac{e^{j\left[\omega t - k\sqrt{x^2+y^2+z^2}\right]}}{\sqrt{x^2+y^2+z^2}}$$

$$k = \frac{\omega}{c} n$$



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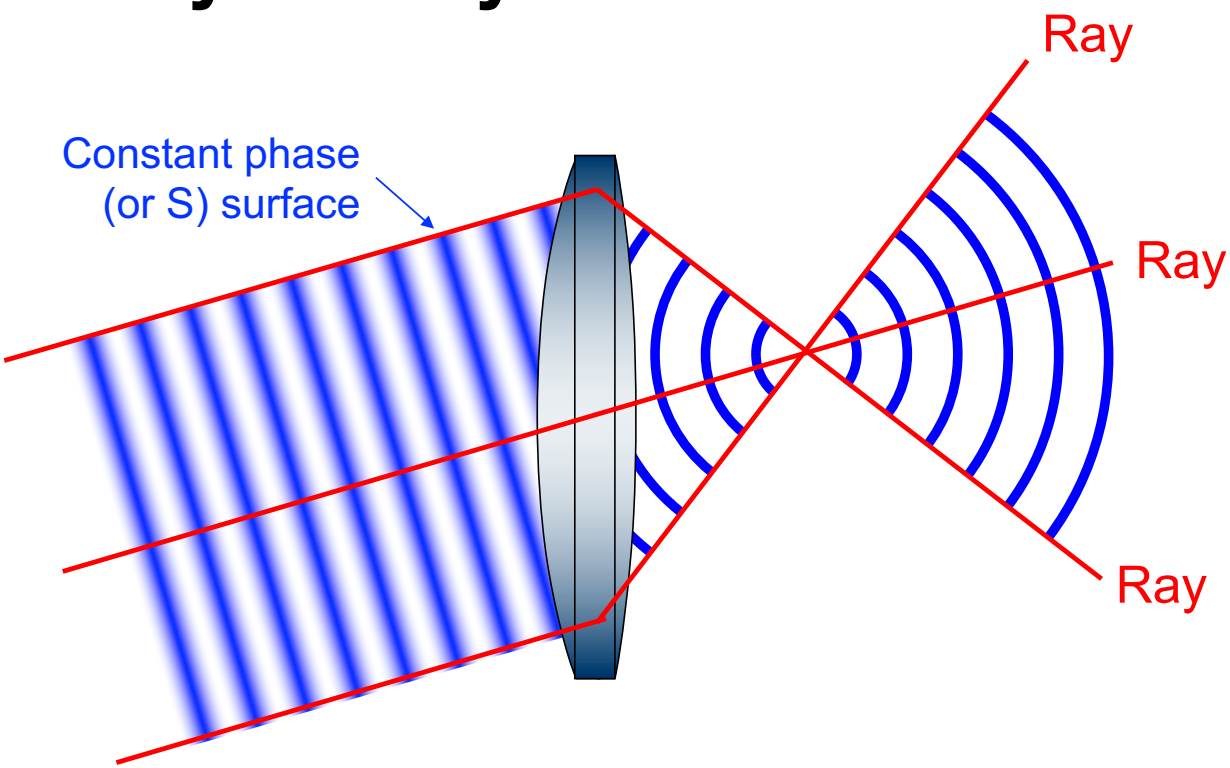
Why use rays?



The electric field $E(\vec{r})$ is a complex function of space.

Rays are lines. ✓

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